Neutron spectroscopy measurements of soft matter (films)

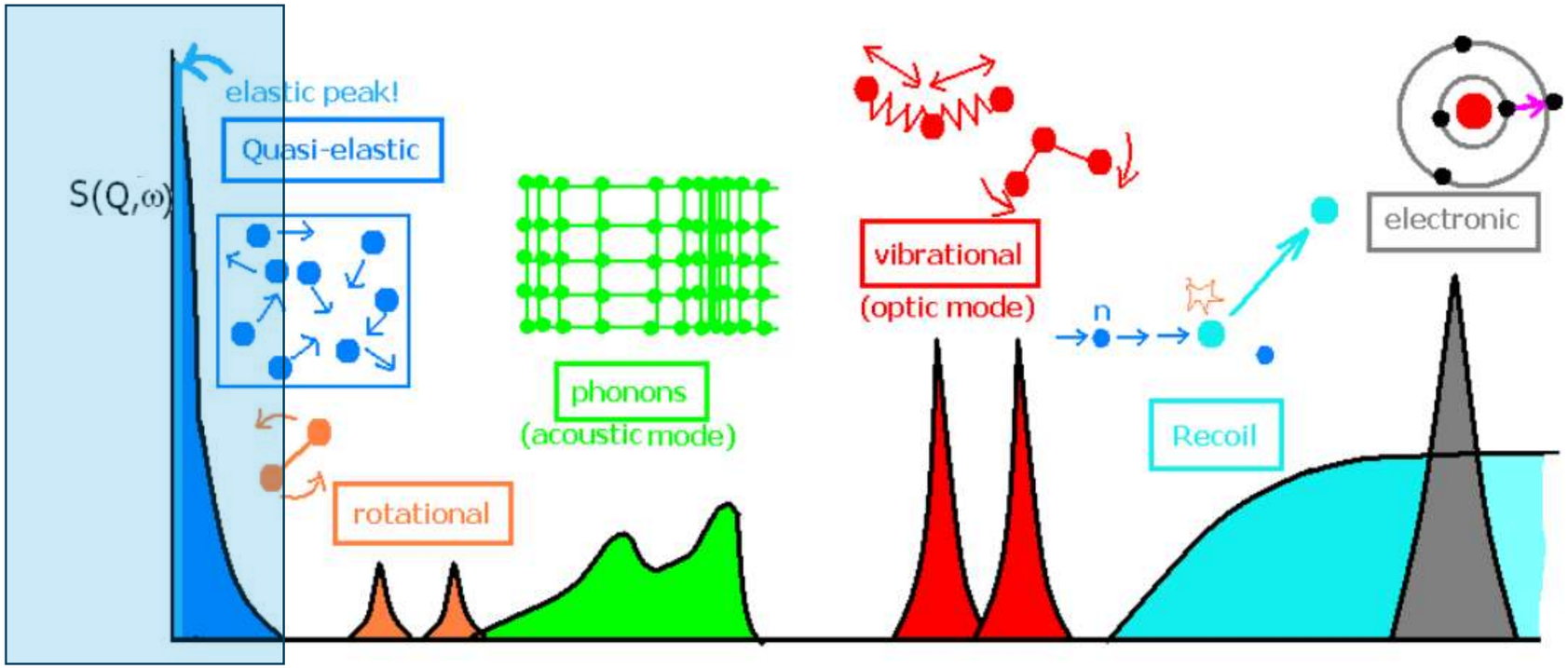
PATCHAVIT JANEWITHAYAPUN (LEO)

What, why, when?

What is neutron spectroscopy?

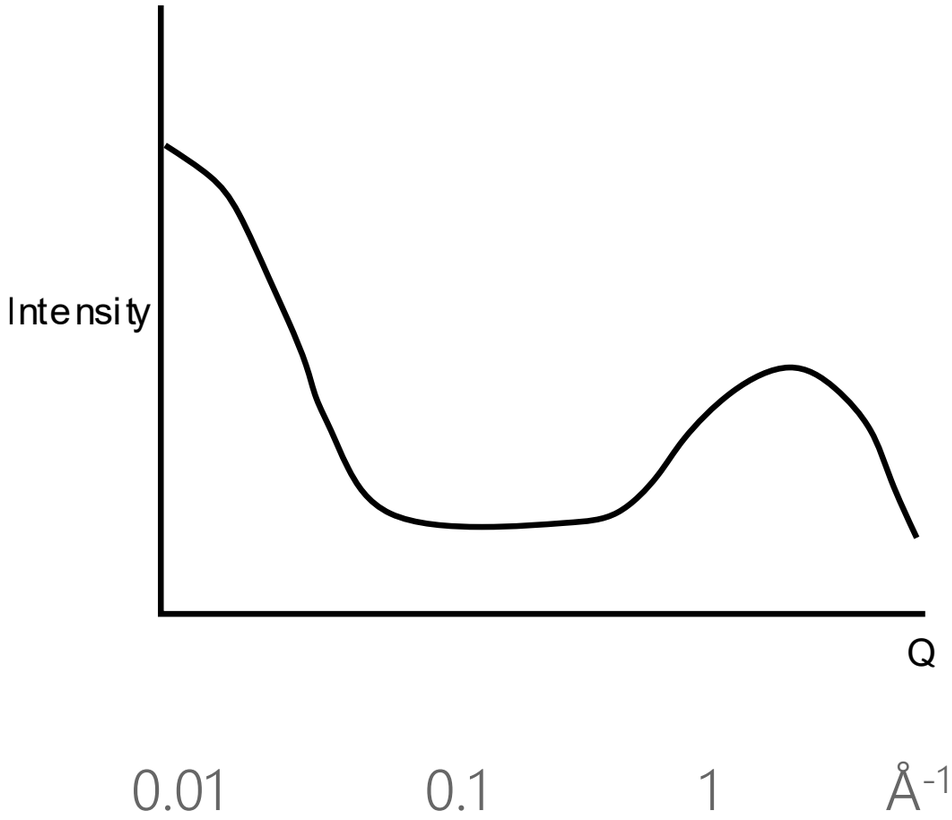
Spectroscopy, study of the absorption and emission (+transmit or reflect) of light and other radiation by matter, as related to the dependence of these processes on the wavelength of the radiation.

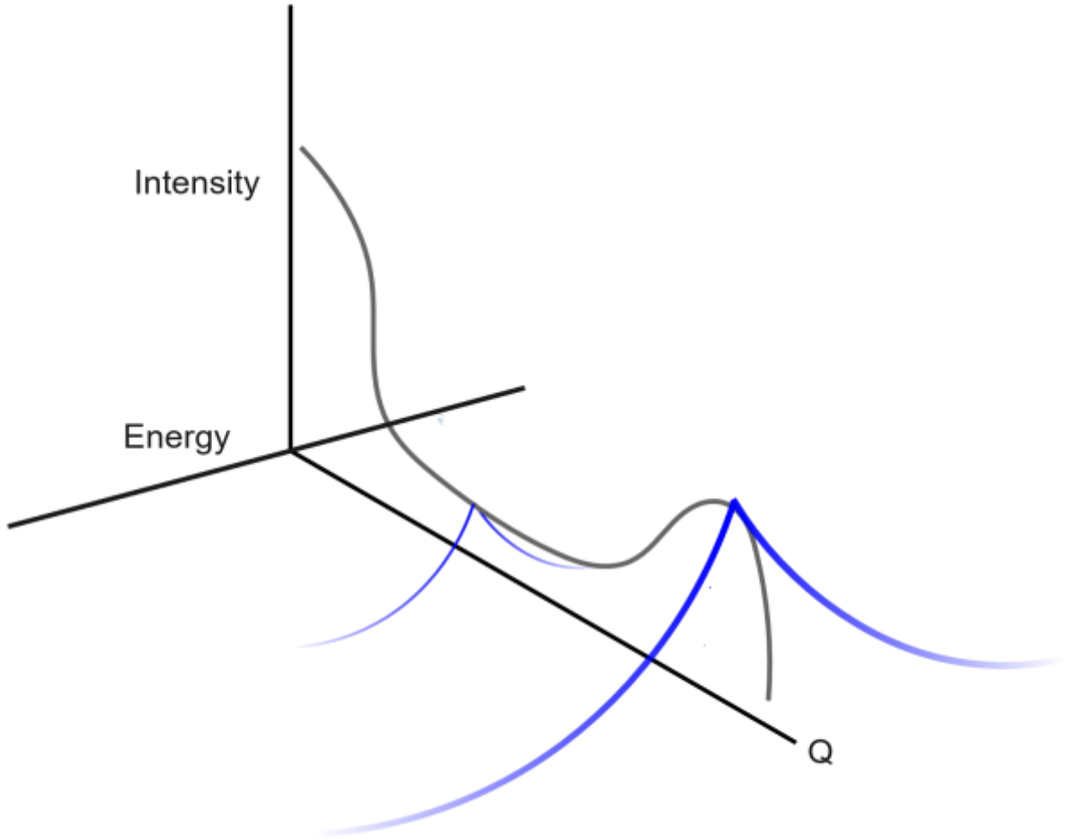
https://www.britannica.com/science/spectroscopy
https://science.nasa.gov/mission/webb/science-overview/science-explainers/spectroscopy-101-introduction/



https://www.oxfordneutronschool.org/2017/Lectures/OSNS17-VGarciaSakai-QENS.pdf

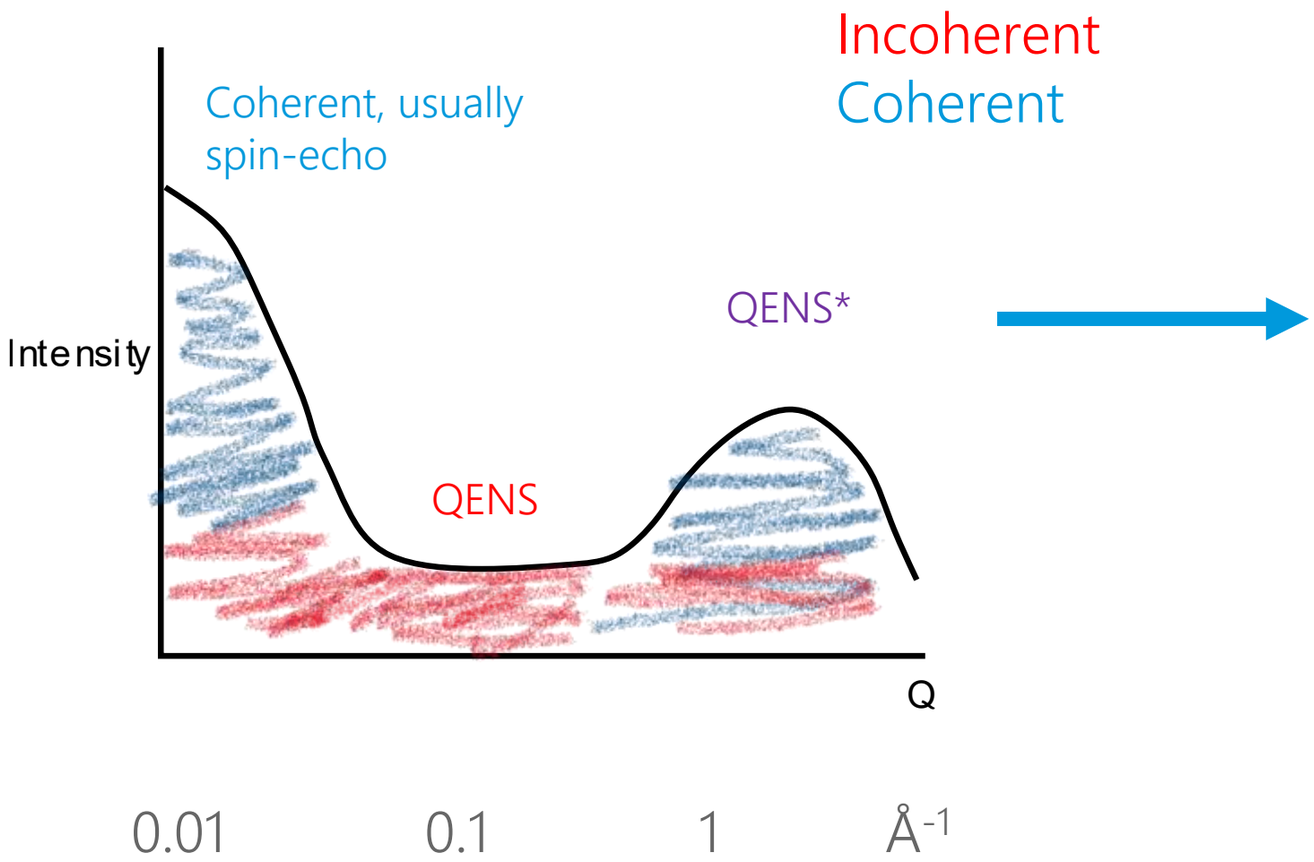
S(Q,E) or S(Q, $\hbar\omega$) or S(Q,ω)

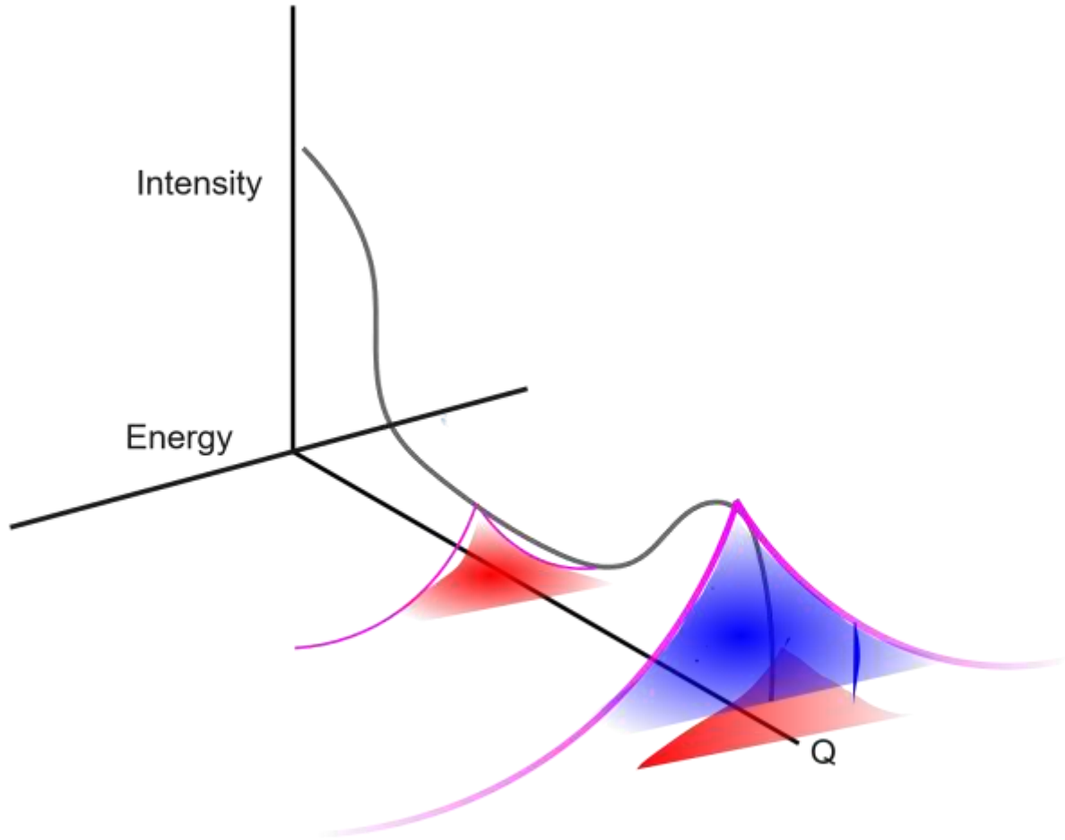




↑ Width of S(Q,E) -> ↑ frequency
Change with Q -> non-localized motion

Some technicalities





What can we do with QENS of soft matter (films)

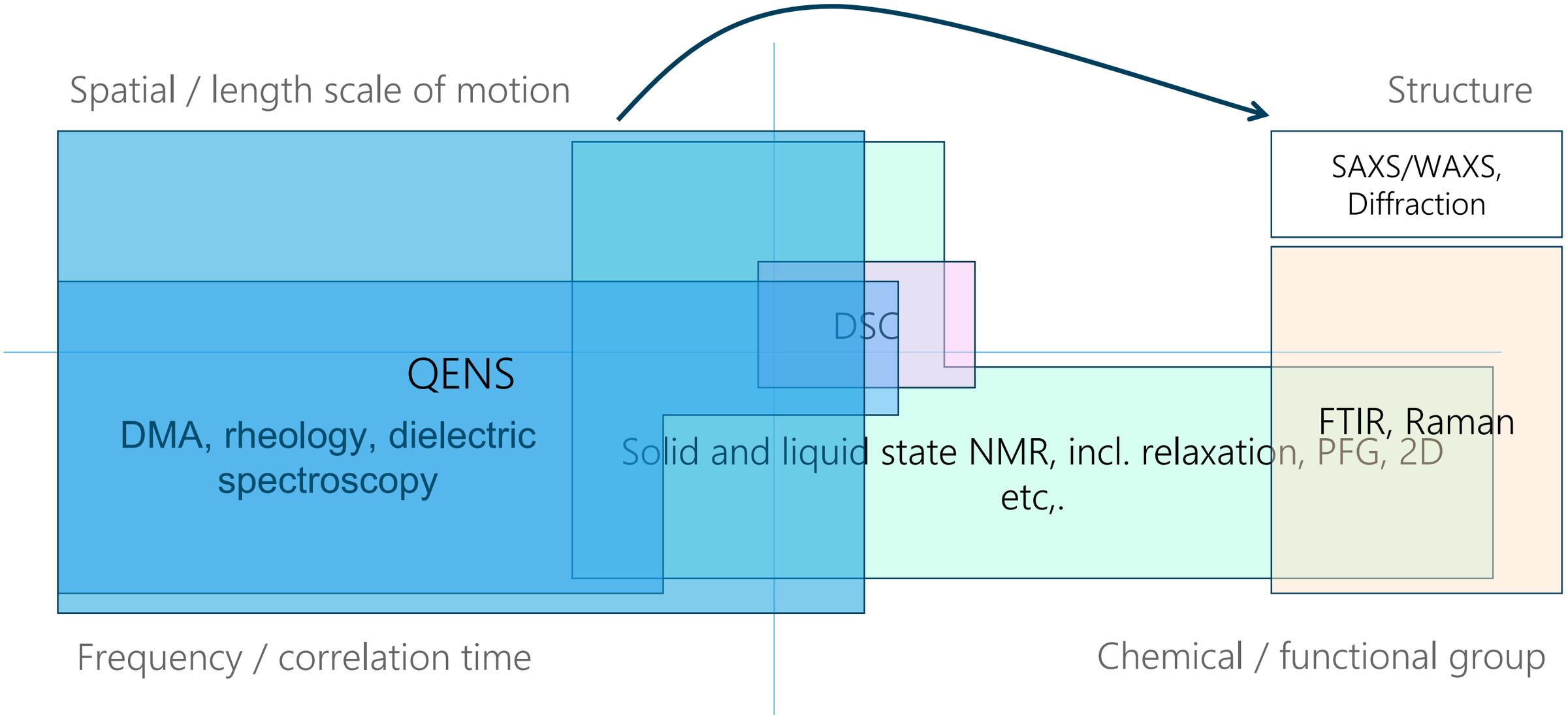
A lot of confined motions

- Films
- Powders
- Fibers
- Foams
- Gels

More diffusive and rotational

- Solutions
- Liquids
- Dispersions
- Melts

What kind of information do we obtain



What is QENS good at?

- **Strongly sensitive to H, so good signals for soft matter and water**
- **Extracting (by fitting) values for correlation times, and population of mobile components**
- **Q-dependence of motions shows whether they are confined in space, or more freely diffusing**
- Sensitive to ps-ns motions, e.g. side groups and segmental motions
- Can characterize bound and free water
- Replacing H_2O with D_2O can help identify water contributions (to an extent)
- Rather flexible sample shape and form, e.g. not necessarily complete well shaped film
- Relatively wide range of sample environments: temperature, humidity, pressure. Efforts to expand this.
- Possibilities for simultaneous diffraction at some instruments
- Relatively well supported by facilities and community: software, seminars, (access*)

Example 1: Modified arabinoxylan and thermal relaxations

- I want to observe molecular relaxations in my material

  - DSC (fast)
  - NMR
  - QENS
  - Dielectric
  - DMA

- I want to observe them as a function of temperature

  - DSC (fast)
  - NMR*
  - QENS
  - Dielectric
  - DMA

- I am quite strongly interested in minor (sub-glass) relaxations

  - QENS
  - NMR*
  - DMA
  - Dielectric

- Practical limitations

  Some of my films were too brittle for DMA
  Limited access to dielectric and NMR resources
  Transitions in polysaccharides are not strong on DSC

- **QENS**
- NMR
- Some DMA

Example 2: Silk fibroin, moisture, temperature and structural evolution

- I want to observe molecular relaxations in my material

  - DSC (fast)
  - NMR
  - QENS
  - Dielectric
  - DMA

- I want to observe them as a function of temperature

  - DSC (fast)
  - NMR*
  - QENS
  - Dielectric
  - DMA

- My material crystallizes during this process

  - QENS (Diff)
  - NMR
  - DMA*
  - Dielectric*
  - FTIR
  - WAXS

- I am also interested in water behavior

  - TGA
  - NMR
  - DSC
  - QENS
  - Vapor sorption

- QENS - DSC (fast)

- DMA - FTIR

Practical tips for experiments

Which instrument to choose

Main criteria: Energy resolution and kinematic range

Backscattering

IN16b (ILL)
HFBS (NIST)

Energy resolution: ~<1 μeV
Kinematic range: ~ -30 μeV to 30 μeV

Good for
Segmental dynamics

Ok for
Side group dynamics << T_g
Water dynamics < 273 K

Indirect geometry

OSIRIS/IRIS (ISIS neutron source)
IN16b BATS mode* (ILL)
BASIS (SNS)
DNA (Jparc)
Miracles (ESS)**

Energy resolution: ~2 μeV to 20 μeV
Kinematic range: ~ ±0.1 meV to ±2 meV

Good for
Water dynamics > 273 K

Ok for
Side group dynamics ~ T_g
Segmental dynamics > T_g

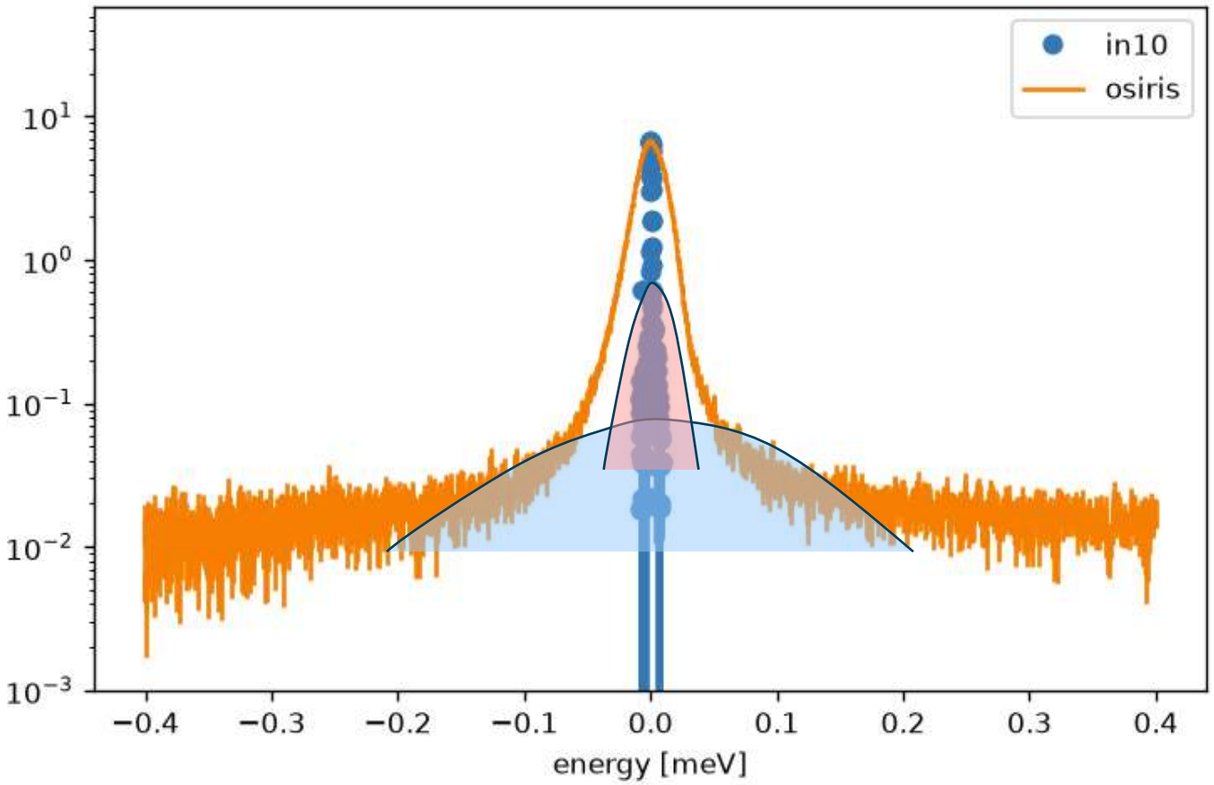
Direct geometry

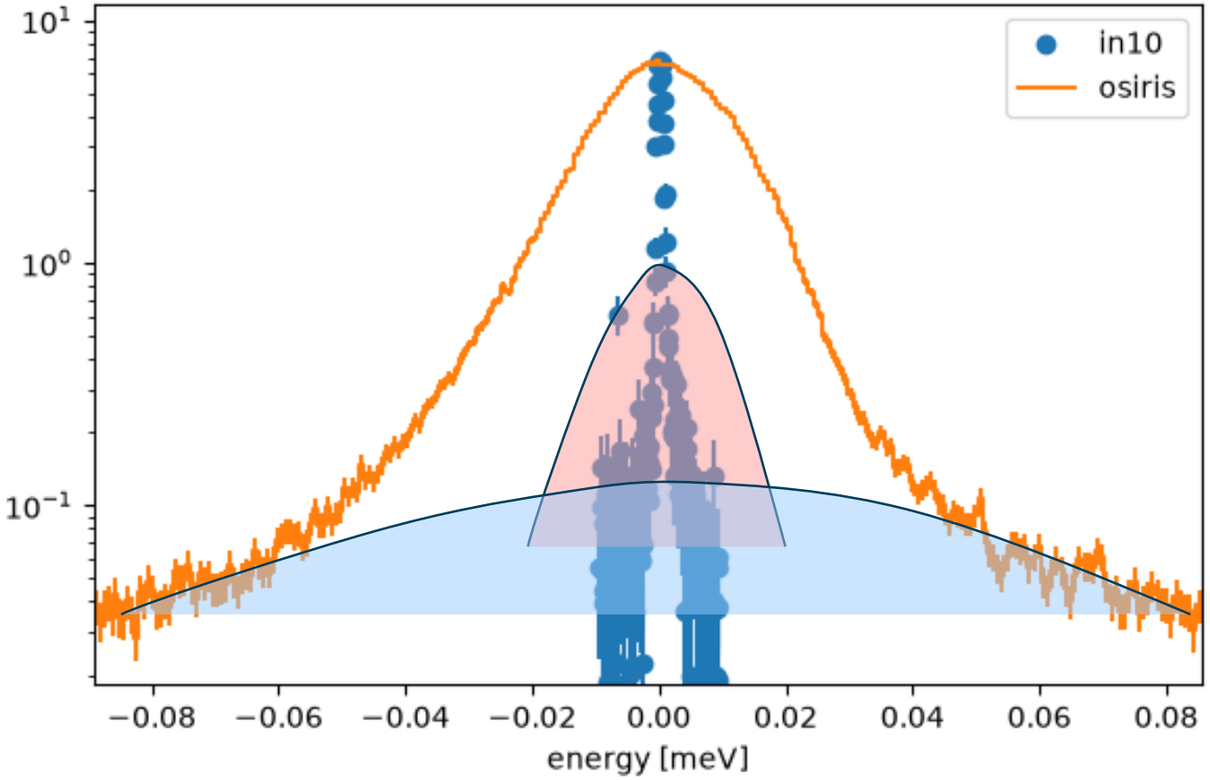
IN5 (ILL)
SHARPER (ILL)
FOCUS (PSI)
AMATERAS (Jparc)
CSPEC (ESS)**

Energy resolution: ~50 μeV to 1 meV
Kinematic range: ~ ±2 meV to ±10s meV

Good/Ok for
Water dynamics > 273 K
Side group dynamics > T_g
Segmental dynamics >> T_g

Which instrument to choose

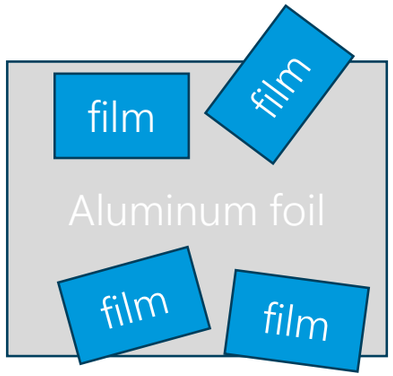




Sample and sample environments

Sample size: ~4-5 cm high
Wide enough to wrap around can ~5 cm
Thickness to give 90% transmission at 5 Å (0.1-0.2 mm)
https://www.ncnr.nist.gov/resources/activation/





Wrap, crimp, roll

Environments

Standard:

Cryostat/cryofurnace: Temperature control from 2 K to ~500 K

Depends on facility:

Pressurized cell

Humidity chamber

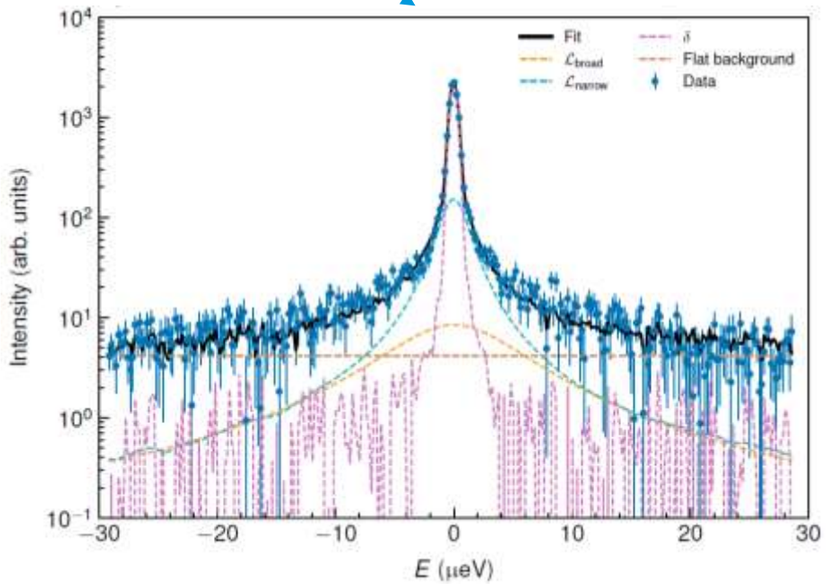
Cell with light exposing option

Credit: STFC online QENS course
https://lmsweb.stfc.ac.uk/moodle/mod/lesson/view.php?id=3092&pageid=703

Process from idea to reduced data

1. Idea + sample + some preliminary data
2. Write proposal 1st draft
3. Get feedback from PI / supervisor etc,.
4. **Discuss with beamline scientist.**
5. Finalize proposal, submit
6. Perform experiment, **QENS measurements take ~2-4 hrs for a S(Q,E)**. For 3 temps/sample, a sample change might be every 12 h
7. You need a resolution (pure elastic scatterer). In the sample can you use, either vanadium or low temp. of sample. You also need an empty can.
8. **Ideally, you do all data reduction before leaving, or very soon after experiment**

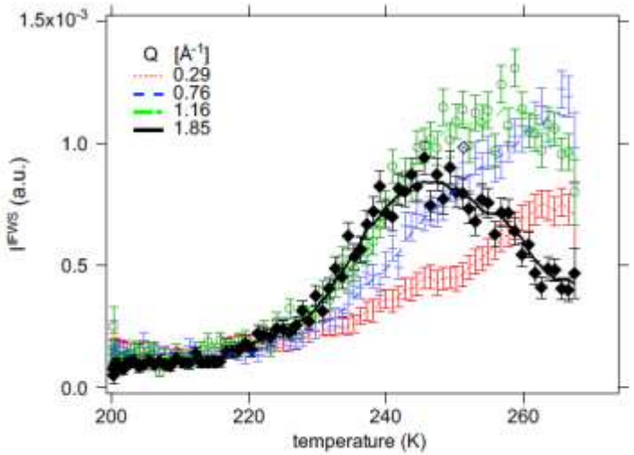
Fixed window scans and/or full spectra measurements



- Ideally measure both
- They can be analyzed simultaneously, adding much more information to fits

- Reponse of mobility vs. parameter
- Shows how many transitions
- Can see Q-dependence

- But more difficult to get correlation time/frequency
- More difficult to get confinement



Tools for data analysis, useful links

Common software for data analysis

1. EasyDynamics (ESS)
   - https://easyscience.github.io/dynamics-lib/latest/
   - https://github.com/ess-dmsc-dram/dmsc-school
2. Jscatter
   - https://jscatter.readthedocs.io/en/latest/
3. Mantid
   - https://www.mantidproject.org/

Mostly designed for Python / Jupyter

Both GUI and Python

Older software but still in use at some beamlines

1. LAMP
2. DAVE

Resources for further reading

Applied / Entry level

- https://www.oxfordneutronschool.org/2017/Lectures/OSNS17-VGarciaSakai-QENS.pdf
- https://lmsweb.stfc.ac.uk/moodle/login/index.php look for ISIS courses -> QENS
- https://juser.fz-juelich.de/record/22573/files/D07_Wuttke.pdf
- https://sons.uniroma2.it/lectures/
- A Practical Guide to Quasi-elastic Neutron Scattering
- Grimaldo M, Roosen-Runge F, Zhang F, Schreiber F, Seydel T. Dynamics of proteins in solution. *Quarterly Reviews of Biophysics*. 2019;52:e7.

More theoretical references

- M. Bée, Quasi-elastic Neutron Scattering Principles and Application in Solid State Chemistry, Biology and Materials Science (1988)
- G.L. Squires, Introduction to the Theory of Thermal Neutron Scattering